

## NMTC Problem-Set 1

**Instructions:** The following are the solutions for Problem Set 1. We have included the questions in their entirety, and then provided their solutions, although we may only write the solutions from the next time. Furthermore, these solutions are purposefully verbose, in order to be as clear as possible about the steps. Future problem sets won't be written as explicitly. The sources to all the problems are at the bottom of the page.

## Problem 1: Miscellaneous

Consider a UFO on a number line starting at  $x = 0$ . Every second the UFO moves a constant, unknown distance  $d$  in the direction of the positive x-axis, where  $d$  is a positive integer. For example, at time  $t = 1$ , the UFO will be at the coordinate  $x = d$ , at time  $t = 2$ , it will be at the coordinate  $x = 2d$ , and so on.

At any time  $t$ , you can check a random point on the number line, and see if the UFO is there or not. Prove that you can always 'catch' the UFO after a finite amount of time.

(Hint: There may be more than 1 answer)

*Proof.* Note first, that if we let  $a_t$  be the coordinate of the UFO at an arbitrary time  $t$ , then  $a_t = td$ . We let  $b_t$  be the coordinate that we check at time  $t$ . Our goal is then to find a value of  $t$  such that  $a_t = b_t$ .

1. The first solution proceeds as follows. Let  $b_t = t^2$ . Then, at time  $t = d$ , we have  $a_d = d^2$ , and  $b_d = d^2$ , so we catch the UFO at time  $t=d$ .
2. For the second solution, we instead define  $b_t = \frac{t(t+1)}{2}$ . Equating  $a_t$  and  $b_t$ , we see that we catch the UFO when  $t = 2d - 1$ .

□

## Problem 2: Algebraic Manipulations

- (a) There is a unique positive real number  $x$  such that the three numbers  $\log_8(2x)$ ,  $\log_4(x)$ , and  $\log_2(x)$ , in that order, form a geometric progression with positive common ratio. Find  $x$ .
- (b) Let  $x, y$ , and  $z$  be real numbers satisfying

$$\log_2(xyz - 3 + \log_5 x) = 5,$$

$$\log_3(xyz - 3 + \log_5 y) = 4,$$

$$\log_4(xyz - 3 + \log_5 z) = 4.$$

Find the value of

$$|\log_5 x| + |\log_5 y| + |\log_5 z|.$$

- (c) Let  $a > b$  be positive integers. Find the sum of all numbers which can be expressed in the form  $\frac{1}{2^a 5^b}$ .

*Proof.* (a) Let the common ratio be  $r$ . We have:

$$\frac{\log_2(x)}{\log_4(x)} = r \iff r = 2$$

Now, we have:

$$\begin{aligned} \frac{\log_4(x)}{\log_8(2x)} &= 2 \\ \iff \frac{\log_2(x)}{\log_2(4)} &= 2 \frac{\log_2(2x)}{\log_2(8)} \\ \iff \log_2(x^3) &= \log_2(16x^4) \\ \iff \frac{1}{16} &= x \end{aligned}$$

And we're done.

(b) We have:

$$\begin{aligned}xyz - 3 + \log_5 x &= 32, \\xyz - 3 + \log_5 y &= 81, \\xyz - 3 + \log_5 z &= 256.\end{aligned}$$

Summing these equations we have:

$$3xyz + \log_5(xyz) = 378$$

From here, we guess(!) that  $xyz = 125$ . Since this works, and the *LHS* is strictly increasing, we know this is the only solution. Now we can find that  $\log_5 x = -90, \log_5 y = -41, \log_5 z = 134$ . The answer is hence,  $90 + 41 + 134 = 265$

(c) Writing out the numbers, we have:

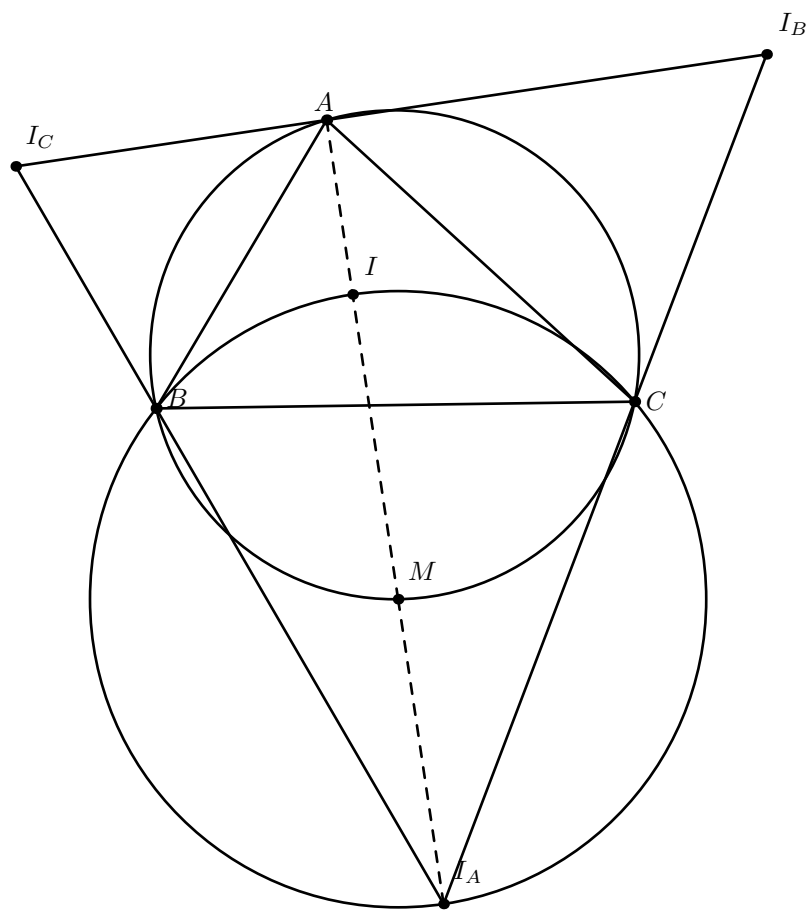
$$\begin{aligned}& \frac{1}{5} \left( \frac{1}{2^2} + \frac{1}{2^3} + \cdots \right) + \frac{1}{5^2} \left( \frac{1}{2^3} + \frac{1}{2^4} + \cdots \right) \\&= \sum_{i=1}^{\infty} \frac{1}{5^i} \left( \frac{1}{2^{i+1}} + \frac{1}{2^{i+2}} + \cdots \right) \\&= \sum_{i=1}^{\infty} \frac{1}{5^i} \left( \frac{1}{2^i} \right) \\&= \sum_{i=1}^{\infty} \frac{1}{10^i} \\&= \frac{1}{9}\end{aligned}$$

□

### Problem 3: Angle Chasing

- (a) In acute  $\triangle ABC$  with incenter  $I$ , prove that  $\angle BIC = 90^\circ + \frac{1}{2}\angle BAC$
- (b) In acute  $\triangle ABC$  with incenter  $I$ . Let  $M$  be the midpoint of the arc  $BC$  of circle  $ABC$  which does not contain  $A$ . Show that  $M$  is the circumcenter of  $\triangle BIC$
- (c) In acute triangle  $\triangle ABC$ , let  $I_A$  be the reflection of the incenter  $I$  over  $M$  (Defined as in the previous problem). Show that  $BI_A$  and  $CI_B$  are the external angle-bisector of  $\angle ABC$  and  $\angle ACB$  respectively.
- (d) In acute  $\triangle ABC$ , Define  $I_B$  and  $I_C$  similarly to how  $I_A$  was defined earlier. Show that  $I$  is the orthocenter of  $\triangle I_AI_BI_C$  (\*)

The diagram on the next page will be referred to throughout the solution.  
The page after that contains the proofs.



*Proof.* (a) We have:

$$\begin{aligned}\angle BIC &= 180^\circ - (\angle IBC + \angle ICB) \\ &= 90^\circ + \frac{1}{2}(180^\circ - \angle ABC - \angle ACB) \\ &= 90^\circ + \frac{1}{2}\angle BAC\end{aligned}$$

as needed.

- (b) We know  $MB = MC$ . Hence, it suffices to show  $MB = MI$ . Let  $\angle BAC = 2\alpha$ ,  $\angle ACB = 2\gamma$ , and  $\angle ABC = 2\beta$ . We then have:

$$\angle MAB = \angle MCB = \angle MBC = \alpha = \angle IAB.$$

Hence,  $A, I, M$ , are collinear. By part (a), we get  $\angle AIB = 90^\circ + \gamma$ . We then have:

$$\angle MIB = 180^\circ - \angle AIB = 90^\circ - \gamma = \alpha + \beta = \angle MBC + \angle ICB = \angle IBM$$

Hence,  $\triangle IBM$  is isosceles and it follows  $MB = MI$ , as needed.

- (c) Note that since  $II_A$  is the diameter of the circle with center  $M$ , we have  $\angle I_ABI = 90^\circ$ . Using the notation of the previous solution, we know that the exterior angle of  $\angle ABC$  is given by:

$$\begin{aligned}180^\circ - 2\beta &= 2(90^\circ - \beta) \\ &= 2(\angle I_ABI - \angle CBI) \\ &= 2(\angle I_ABC)\end{aligned}$$

Hence,  $I_AB$  is the exterior angle bisector of  $\angle ABC$  and similarly for  $I_AC$ .

- (d) By part (c) and (b), we have that:

$$\begin{aligned}\angle I_AAI_B &= \angle I_AAC + \angle I_BAC \\ &= \angle IAC + \angle I_BAC \\ &= \alpha + \frac{1}{2}(180^\circ - 2\alpha) \\ &= 90^\circ\end{aligned}$$

Similarly,  $\angle I_A A I_C = 90^\circ$ . Hence,  $I_B, A, I_C$  are collinear. It follows that  $I_A A$  is an altitude of  $\triangle I_A I_B I_C$ . By symmetry,  $I_B B$ , and  $I_C C$  are altitudes as well and they concur at  $I$ , proving that it must be the orthocenter.

□



**Problem 4: Messy Algebra?**

- (a) Let  $m$ , and  $n$  be non-negative integers such that  $n \geq m$  Prove that:

$$\sum_{k=0}^m \binom{m}{m-k} \binom{n}{k} = \binom{m+n}{m}$$

- (b)

$$\binom{\binom{n}{2}}{2} = 3 \binom{n+1}{4}$$

Prove that, for all  $n \geq 3$ , the above equality holds.

For both problems, there's an obvious method, and a correct method. (\*)

*Proof.* The obvious method for both problems is to expand the binomial coefficients in terms of their definitions and then try to cancel everything out. We won't be showing that here, simply because it's completely uninformative. Instead, consider the following.

- (a) Notice that the right hand side is simply the number of ways to choose  $m$  elements from a collection of  $m+n$  elements.

On the other hand, consider "splitting" the  $m+n$  objects into a collection of  $m$  objects, and a collection of  $n$  objects. For any given  $k$ , with  $0 \leq k \leq m$ , the terms inside the sum in the left hand side counts how many ways there are to choose  $m-k$  objects from the smaller set of  $m$  objects, and SIMULTANEOUSLY(hence the multiplication) choose  $k$  objects from the set of  $n$  objects. For each value of  $k$ , we are choosing  $m$  objects. Summing over all possible values of  $k$  gives the total number of ways to choose  $m$  objects from a set of  $m+n$  objects. Hence, the terms on the *LHS* and *RHS* must be equal.

- (b) Consider the *LHS*. Given  $n$  points on the plane,  $\binom{n}{2}$  is the number of lines you can draw. Then  $\binom{\binom{n}{2}}{2}$  is simply the number of ways to choose 2 lines from these  $\binom{n}{2}$  lines, i.e the number of ways to draw 2 lines given  $n$  points on the plane.

We now count the number of ways to draw 2 lines another way. Notice that you can either draw 2 lines with 3 points, or 2 lines with 4 points. If you take 3 points on the plane, there are 3 ways to draw 2 distinct lines between them (draw them). Similarly, if you take 4 points on the plane, there are also 3 ways to draw 2 distinct lines between them. Thus, the total number of ways to choose 2 lines on the plane is

$$\begin{aligned} &= 3\binom{n}{3} + 3\binom{n}{4} \\ &= 3\left(\binom{n}{3} + \binom{n}{4}\right) \\ &= 3\binom{n+1}{4} \end{aligned}$$

□

**Sources:**

1. Rakin Khurram
2. (a) 2020 AIME I Problem 2  
(b) 2016 AIME II Problem 3  
(c) 2012 NIMO Day 4 Problem 5
3. Incentre-Excentre Lemma
4. (a) Vandermonde's Identity  
(b) Folk-Lore